SpeakerLM: End-to-End Versatile Speaker Diarization and Recognition with Multimodal Large Language Models

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Abstract

The Speaker Diarization and Recognition (SDR) task aims to predict "who spoke when and what" within an audio clip, which is a crucial task in various real-world multi-speaker scenarios such as meeting transcription and dialogue systems. Existing SDR systems typically adopt a cascaded framework, combining multiple modules such as speaker diarization (SD) and automatic speech recognition (ASR). The cascaded systems suffer from several limitations, such as error propagation, difficulty in handling overlapping speech, and lack of joint optimization for exploring the synergy between SD and ASR tasks. To address these limitations, we introduce **SpeakerLM**, a **unified multimodal large language model for SDR** that jointly performs SD and ASR in an **end-to-end** manner. Moreover, to facilitate diverse real-world scenarios, we incorporate a **flexible speaker registration mechanism** into SpeakerLM, enabling SDR under different speaker registration settings. SpeakerLM is progressively developed with a multi-stage training strategy on large-scale real data. Extensive experiments show that SpeakerLM demonstrates strong data scaling capability and generalizability, outperforming state-of-the-art cascaded baselines on both in-domain and out-of-domain public SDR benchmarks. Furthermore, experimental results show that the proposed speaker registration mechanism effectively ensures robust SDR performance of SpeakerLM across diverse speaker registration conditions and varying numbers of registered speakers.

Code — https://sites.google.com/view/speakerlm/overview

1 Introduction

Speaker Diarization (SD) partitions an audio into homogeneous segments according to speaker identities, effectively answering "who spoke when" (Tranter and Reynolds 2006; Anguera et al. 2012; Park et al. 2022). In real-world scenarios such as multi-speaker meetings, knowing "who spoke when" is insufficient. Associating speaker labels with speech transcripts is more meaningful, as it addresses the more comprehensive question of "who spoke when and what" (Gao et al. 2025). We refer to this task as **Speaker Diarization and Recognition (SDR)**, which jointly performs SD and Automatic Speech Transcription (ASR) for enriched conversational understanding.

Conventional SDR systems typically pipeline an SD module and an ASR module (Raj et al. 2021; Yu et al. 2022; Cornell et al. 2024; Boeddeker et al. 2024). The SD module segments the audio stream into speaker-homogeneous regions, which are then aligned with the ASR outputs, and the speech content for each identified speaker is transcribed. However, such cascaded systems suffer from several limitations. First, error propagation is a major concern. Inaccuracies in SD, such as incorrect speaker boundaries or label assignments, are passed on to the ASR module, leading to degraded transcription quality and incorrect speaker-attributed text (Cornell et al. 2023). In addition, conventional SD systems struggle to handle overlapping speech, which is common in real-world conversations, resulting in limited performance (O'Shaughnessy 2025). Furthermore, the cascaded system lacks joint optimization as ASR and SD modules are usually trained independently with different datasets and frameworks. The disjoint training paradigm prevents the SDR system from fully leveraging the synergy between SD and ASR tasks, limiting the overall SDR performance.

Two main paradigms have been attempted to address these limitations of cascaded SDR systems. One paradigm, namely Speaker-Attributed ASR (SA-ASR), jointly trains SD and ASR. As a representative end-to-end SA-ASR system, SA-Transformer (Liang et al. 2023) incorporates an ASR encoder and a speaker encoder, and achieves competitive SDR results. However, SA-ASR systems can only operate under the condition that pre-extracted speaker embeddings are available. Another paradigm uses Large Language Models (LLMs) (Wei et al. 2022; Chang et al. 2024) for post-processing. Due to the disjoint training issue in cascaded SDR systems, temporal mismatches between the two modules may lead to word-level diarization errors (Wang et al. 2024). Prior works (Park et al. 2024; Efstathiadis, Yadav, and Abbas 2025) explore the strong generalization and reasoning abilities of LLMs to post-process the outputs of a cascaded SDR system, refining speaker attributions and correcting transcription inconsistencies. While such post-hoc adjustments may improve the overall SDR performance, they are inherently limited by the quality of the initial SD and ASR outputs, and cannot fully resolve issues arising from upstream error propagation or misalignment between

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modalities. Therefore, how to leverage LLMs not just for post-processing but as core components in end-to-end SDR systems remains an open challenge.

To address the limitations of all these prior works, we introduce **SpeakerLM**, a multimodal LLM (MLLM) specially designed for end-to-end SDR. Furthermore, to better handle real-world complex multi-speaker conversational scenarios, we incorporate a flexible speaker registration mechanism into SpeakerLM, enabling SpeakerLM to versatilely adapt to varying levels of speaker information availability in real-world applications. We propose three different speaker registration conditions, including cases where prior speaker identities are unknown, known, and redundant.

Our contributions can be summarized as follows:

- **The first MLLM for end-to-end SDR.** To the best of our knowledge, our SpeakerLM is the first MLLM designed for end-to-end SDR. Specifically, we apply an audio encoder and two projectors as the front-end, leading to an encoder-projector-LLM architecture tailored for SDR.
- **A flexible speaker registration mechanism for versatile SDR.** We incorporate a flexible speaker registration mechanism into SpeakerLM, where the prior speaker embeddings are projected and concatenated with audio and text tokens, enabling the model to handle diverse multi-speaker scenarios in real-world applications.
- **Extensive analysis.** To evaluate the effectiveness of SpeakerLM, we build various cascaded SDR baselines. Results show that SpeakerLM demonstrates strong data scaling capability, outperforming all the baselines on both in-domain and out-of-domain public benchmarks.

2 Related Work

2.1 Speaker Diarization

Traditional speaker diarization frameworks typically consist of multiple components, including Voice Activity Detection (VAD) to identify speech segments, speaker embedding extraction to represent speaker characteristics, and clustering to group speaker embeddings for identifying speakers. 3D-Speaker (Chen et al. 2025b) is a representative traditional SD system following this pipeline, and achieves competitive performance across multiple public benchmarks.

Conventional SD systems struggle with overlapping speech, because VAD is designed under the assumption of single-speaker segments. One promising solution to address this issue is to replace the VAD module with a neural speaker activity detection module, which predicts active speakers in each time frame (Fujita et al. 2019; Kinoshita, Delcroix, and Tawara 2021b,a). Along this direction, Pyannote (Bredin et al. 2020; Bredin 2023) achieves strong performance on different datasets. Based on Pyannote, Diarizen (Han et al. 2025a,b) incorporates representations from the pretrained WavLM model (Chen et al. 2022) into the neural speaker activity detection module, outperforming Pyannote on various datasets by providing richer acoustic information.

2.2 Speaker Diarization and Recognition

Figure 1 (a) depicts a conventional SDR system that pipelines SD and ASR modules (Cornell et al. 2024), denoted as **SD+ASR**. This pipeline has several limitations, such as error propagation and lack of joint optimization. To address these issues, one direction uses LLMs for post-processing, and another direction jointly models SD and ASR tasks within a single unified model in an end-to-end manner. As shown in Figure 1, we denote these two methods by **SD+ASR+LLM** and **E2E-SDR**, respectively.

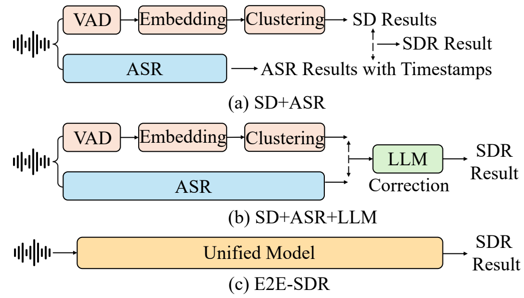

Figure 1: Comparison between three different SDR frameworks: (a) SD+ASR (b) SD+ASR+LLM and (c) E2E-SDR.

For E2E-SDR, existing studies typically incorporate pre-extracted speaker embeddings for all speakers. This framework, also referred to as SA-ASR (Liang et al. 2023), assumes that the provided speaker embeddings exactly match the speakers in the ground truth. Consequently, it struggles to handle mismatched scenarios such as unregistered speakers or over-registered cases. Our work follows the E2E-SDR paradigm, but is different from the prior SA-ASR systems. Specifically, we introduce a more flexible speaker registration mechanism during training, enabling the model to handle speaker registration inconsistencies effectively. Compared with the cascaded frameworks, our unified modeling approach allows for exploring the synergy between SD and ASR tasks and joint optimization, leading to better alignment between speaker and textual information.

2.3 Multimodal Large Language Model

Recently, integrating audio and text with MLLMs has led to notable advances in auditory comprehension and reasoning. Representative models, such as MinMo (Chen et al. 2025a), Qwen2-Audio (Chu et al. 2024) and Kimi-Audio (Ding et al. 2025), have demonstrated strong capabilities in audio-text understanding and generation tasks. However, most existing MLLMs primarily focus on single-speaker scenarios, where they exhibit strong performance in ASR tasks but leave SD tasks unexplored. More recently, the Multi-Talker LLM (MT-LLM) has been proposed to handle multi-speaker ASR (Meng et al. 2025). However, MT-LLM is limited to detecting speaker change points and do not explicitly tackle speaker attribution within multi-speaker transcripts.

Based on our review of the literature, SpeakerLM is the first audio-text MLLM capable of performing SDR, integrating both speech content understanding and speaker-aware modeling within a unified framework. Unlike prior works, SpeakerLM goes beyond basic ASR or change-point detection and explicitly explores the performance of speaker assignment in multi-speaker transcription tasks.

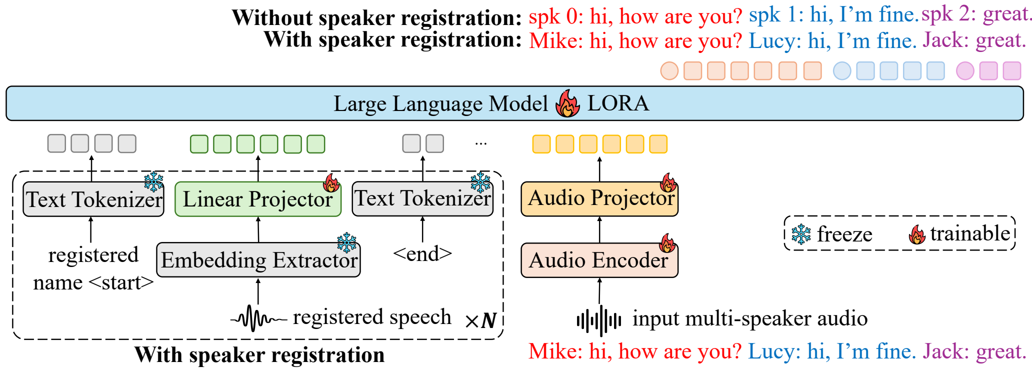

Figure 2: Overall architecture of the proposed **SpeakerLM**. When speaker registration is not performed, each speaker in the transcript is identified by an anonymous ID. With speaker registration enabled, speakers are labeled by their actual names.

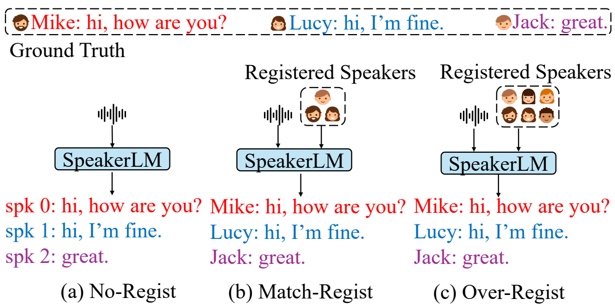

Figure 3: Overview of the proposed three different speaker registration mechanisms: (a) No-Regist (no speaker registration) (b) Match-Regist (exact actual speakers in the audio are pre-registered) (c) Over-Regist (more speakers are registered than actual speakers in the audio). For Match- and Over-Regist, speakers are registered in a random order.

3 SpeakerLM

3.1 Model Architecture

Figure 2 illustrates the overall architecture of SpeakerLM. Specifically, SpeakerLM integrates a *lightweight* modality alignment mechanism into a pre-trained text LLM. For input multi-speaker audio, we use an audio encoder for encoding, followed by a projector to inject the audio embeddings into the feature space of the text LLM. For speaker registration, a frozen text tokenizer is used to tokenize both the registered speaker's name and the special markers (i.e., `<start>` and `<end>`). The registered speaker's speech is first processed by a frozen pre-trained embedding extractor to obtain speaker embeddings, which are then projected by a single linear layer into the LLM backbone.

Audio Encoder and Projector. The audio encoder is initialized with the pre-trained SenseVoice-large encoder (An et al. 2024), which offers robust audio representation capabilities and provides strong performance across various audio understanding tasks such as multilingual speech recognition and audio event detection. For the audio projector, we use a randomly initialized two-layer Transformer (Vaswani et al. 2017), followed by a convolutional neural network layer for dimensional alignment.

Embedding Extractor and Projector. A pre-trained speaker embedding model is utilized to extract speaker embeddings, delivering robust and discriminative representations that are essential for precise speaker identification and attribution. Specifically, we use an open-source speaker embedding model ERes2NetV2 (Chen et al. 2024) for embedding extraction, which has achieved SOTA performance on multiple speaker verification benchmarks. A single linear layer projector is applied to the extracted embedding for dimensional alignment.

Large Language Model. We use the pre-trained Qwen2.5-7B-Instruct model (Team 2024) as the backbone text LLM to benefit from its strong instruction-following and general language understanding capabilities, which allows SpeakerLM to effectively handle complex multi-speaker SDR tasks with different levels of information.

3.2 Flexible Speaker Registration Mechanism

As shown in Figure 2, we incorporate a flexible speaker registration mechanism into SpeakerLM. In order to facilitate real-world scenarios, we propose three different registration strategies, including No-Regist, Match-Regist, Over-Regist, as illustrated in Figure 3. Given the number of speakers in the ground truth as N_{gt} and the number of registered speakers as N_{rg}, the relationship between these two values in different registration settings is formulated as follows:

$$N_{rg} = \begin{cases} 0, & \text{if No-Regist} \\ N_{gt}, & \text{if Match-Regist} \\ N_{gt} + N_{ov}, & \text{if Over-Regist} \end{cases} \tag{1}$$

where $N_{ov} > 0$ is the number of over-registered speakers.

No-Regist means no speaker registration is performed. This is the conventional setting for cascade SD systems and applications. We simply feed multi-speaker audio into the model without providing any prior information about the speakers. This paradigm is align with the traditional cascade SDR framework, where each speaker in the output is represented by a anonymous speaker ID (e.g., spk 0, spk 1, ...).